Ayush Roy

PhD Student, Computer Science and Engineering, State University at New York, Buffalo

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EDUCATION

State University at New York, Buffalo Aug'2024 – Present
Doctor of Philosophy (Ph.D.), Computer Engineering
Advisor: Dr. Vishnu Suresh Lokhande

Jadavpur University, West Bengal Jul'2020 - Aug'2024
Bachelor of Engineering, Electrical Engineering

RESEARCH EXPERIENCE

Lawrence Livermore National Lab, LLNL May'2026 - Aug'2026
Computational Engineering Research Intern

Indian Institute of Scince and Technology, IISc May'2023 - Aug'2023
Summer Research Intern

SERVICES

TMLR, DMLR, ICLR 2027

PUBLICATIONS

See my Google Scholar page for all publications, recent updates and citation information. *(*: equal contributions)*

Peer-Reviewed Conference Proceedings

[C1] A Roy, R Chakraborty, L Varshney, V Lokhande. **Beyond Pooling: Matching for Robust Generalization under Data Heterogeneity.** *AISTATS 2026* [Paper]

[C2] A Roy, S Enam, J Xia, W.H Kim and V.S Lokhande. **Is Exchangeability better than I.I.D to handle Data Distribution Shifts while Pooling Data for Data-scarce Medical image segmentation?** *MIDL 2026* [Paper]

[C3] A Roy, A Lee, R Chakraborty, V.S Lokhande. **ManifoldGD: Training-Free Hierarchical Manifold Guidance for Diffusion-Based Dataset Distillation.** *CVPR 2026* [Paper]

[C4] P Dey*, **A Roy***, H Cho, H.W Kim, V.S Lokhande. **MAGNET: Manifold-Aware Graph Diffusion Network for Connectome Generation.** *NeurIPS 2026* [Paper]

Peer-Reviewed Journals and Magazines

[J1] A Roy*, A.R Kaushik*, V.S Lokhande, N.K Ratha, V Govindaraju. **GE-FM: Geometry-aware Energy-based Flow Matching for Non Euclidean Manifolds.** *TMLR, 2026* [Paper]

Peer-Reviewed Workshop Papers and Extended Abstracts

[W1] A Roy, T Hassan, R Ayyalasomayajula and V.S Lokhande. **Attending To Routers Aids Indoor Wireless Localization.** *AAAI ML4Wireless, 2025* [Paper]

[A1] A Roy, WH Kim, R Zivadinov and V.S Lokhande. **Scanner-aware Causal Harmonization Of ADNI MRI for Alzheimer's Disease Detection.** *AAIC, 2026*

AWARDS

1. NeurIPS 2026 Financial Assistance Award

TEACHING EXPERIENCE

Graduate Teaching Assistant, State University at New York, Buffalo

CSE587: Data Intensive Computing, Fall 2024 *CSE574: Introduction to Machine Learning, Spring 2025* *CSE368: Introduction to Artificial Intelligence, Fall 2025* *CSE573: Computer Vision and Image Processing, Spring 2026* *CSE555: Introduction to Pattern Recognition, Fall 2026*